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Plagiarism Detection and AI Detection Report

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	Text coverage	Words
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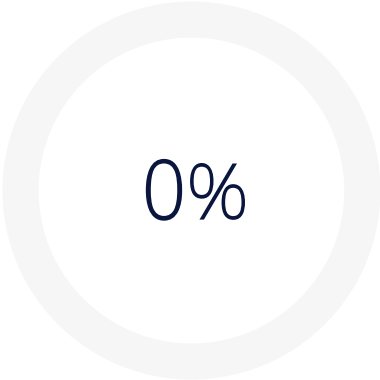
Plagiarism



Results (0)

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Plagiarism types	Text coverage	Words
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Excluded		
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PIE

PATENT INFORMATION EXTRACTION FORM

TOPIC : An Autonomous Neuromorphic Underwater Coral Reef Restoration System with Event Based Detection and Adaptive Indexed Pod Deployment

SCHOOL: Chitkara University Institute of Engineering and Technology

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1. What is the problem?

- The current coral reef monitoring systems are not fully automatic, as they are depend on humans to analyse the data and later take actions, for the coral damage detection and delay the deployment.

- There is a delay between the detection of the coral reef damages and to perform the restoration, which reduces effectiveness and increases cost.

- Existing systems use traditional cameras and cloud processing, which require more energy and are not suitable for real-time response.

- There is no single system that can both detect coral damage instantly and perform restoration at the same location without human involvement

2. How did we solve it (Inventive Step)?

- We designed an autonomous underwater system that can both detect coral damage and respond without human involvement using the help of SNN algorithm

- The system uses an event based sensor which capture change in brightness at each pixel and helps in reducing power usage.

- It processes the data directly on the device, so there is no need for cloud-based analysis.

- When damage is detected, the system automatically deploys coral restoration pods at the exact location.

- A controlled mechanism helps in the accurate placement and avoid repetition of the process in the same area which makes the system more efficient and reliable.

3. What were the other possible solutions and why they could not be done?

- Several other solutions were consider for coral reef monitoring, damage detection and restoration.

- Another option was using normal cameras enabled AI models and then sending data to the server but it needs high bandwidth, more power with the help of CNN based algorithm which is not suitable for long underwater use.

- Manual inspection by divers was also considered but it is slow, expensive and not practical for covering large coral areas.

- Some basic under water robots can place coral at fixed intervals but they cannot detect damage properly and it will lead to the waste of resources.

- Hydraulic systems for deployment were another option but they are complex, heavy and have risks like leakage and high maintenance.

- Because of these issues we chose a smarter system that can detect damage and revive coral reef in real time with an efficient and automatic way.

4. What are the benifits of the solution proposed by us?

- **Real Time Autonomous Operation** : The system detects damage in the coral reef and immediately take action on the same location without waiting for any external analysis.

- **Energy Efficiency**: It used the event based sensor to send the data to the neural processing unit, which with the help of SNN algorithm leads to the reduced power consumption as compared to the traditional camera systems.

- **Reduced Delay**: Since the entire process is happening on the device itself, there is no need to send data to servers, which makes the response time much faster and hence decreases latency.

- **Adaptive Deployment Mechanism** : The system can place restoration pods accurately based on how severe the coral reef is damaged by using XY electric gantry (printer works mechanism) , ensuring better results.

- **Scalability**: It works continuously over a large coral reef area without the help of scuba divers, making it suitable for large scale use.

- **Improved Accuracy**: The system remembers the previous locations and avoids replacing the pods again in the same spot hence reducing the waste of resources.

- **Mechanical Reliability**: The electric mechanism is simpler, reliable and requires less maintenance compared to the ecomplex hydraulic systems.

- **Bandwidth Optimisation**: It processes only the important changes instead of full images which leads to reduced data usage and communication needs.

- **Integrated System:** Unlike the conventional systems used before that separate monitoring and restoration the invention combines both into a unified autonomous platform.

5. Explain the step wise working of the innovation explaining all the components and the specific function they are performing.

- **Step 1 : Underwater deployment and mobility:** The device is deployed underwater and then it navigates to the targeted coral reef area with the help of a thruster based propulsion.

Components used:

- Waterproof pressure sealed enclosure
- Thrusters (vectored fans)
- ESC drivers
- Inertial Measurement Unit Sensor
- Depth sensor
- Battery pack

Function: It provides a stable underwater movement, depth holding and electrical power distribution.

- **Step 2 : Real time reef scanning using event driven camera :**The device scans the coral surfaces continuously with the help of an event driven camera. Only pixel level intensity changes are captured thus reducing the data load.

Components used:

- Event driven camera
- Dual LED illumination
- Optical lens and mounting frame

Function: It helps to identify changes in the texture of the coral reef, death of living organisms, and less biological activity with less power consumption and less delay.

- **Step 3: Coral spawning and egg detection mode:** When the system enters a spawning window or detects a local particle increase in the coral reef, it activates the spawning and egg detection module.

Components used:

- Micro camera
- LED
- Turbidity Sensor
- Particle Sensor
- Micro pump and micro filter sampling chamber

Function: It detects the presence of coral gamete bundles and larvae with the help of optical classification and particle change confirmation, and logs spawning locations for research and future restoration planning.

- **Step 4: Neuromorphic processing and coral reef health classification :** The event stream and egg sensor data is processed locally using a Spiking Neural Network (SNN) running on the onboard processor.

Components used:

- SNN or neuromorphic processing unit AI chip
- Embedded MCU ESP32 or STM32 for control
- Local storage

Function: It classifies the coral reef health as **Healthy, Damaged or Decomposed** and generates a confidence score for the decision making.

- **Step 5: Decision logic and safety checks:** When the coral reef is classified as decomposed or damaged, the microcontroller performs controlled check, before deploying the pods.

Decision conditions include:

- Confidence score must be above the threshold
- Stable hovering (IMU and depth confirmation)

- No obstacle and jam detection
- Treated location verification (memory)

Components used:

- MCU firmware logic
- Jam detection sensor (gate/gantry feedback)
- Localisation/mapping module

Function: It prevents the false deployment and duplicate deployment at the same spot.

➢ **Step 6: XY electric gantry indexing of the selected pods:** After the confirmation, the indexed electric gantry position itself to the selected pod coordinates in the 10×10 storage grid.

Components used:

- X axis stepper motor and linear rail
- Y axis stepper motor and linear rail
- Position encoder/limit switches
- Pod storage grid (100 box pods)

Function: It provides an accurate positioning for a controlled pod selection and avoids mechanical misalignment.

➢ **Step 7: Automatic multi pod release (10 pods) :** Once aligned, the system will release **exactly 10 pods** through the dispensing gate.

Components used:

- Servo actuated release gate
- Pod counter sensor (IR/limit/hall)
- Discharge chute/outlet port

Function: It ensures the precise quantity based deployment for the coral reef regeneration while avoiding over the release.

- **Step 8: Logging, mapping, and communication :** The device stores the event data, coral health status, pod count released, and the treated location. If communication is available it transmits summary data to a surface/server.

Components used:

- Storage module (SSD/Flash)
- Communication module (acoustic modem / RF at surface)
- GPS at surface and underwater localisation (as available)

Function: It maintains restoration records, enables revisit planning, and supports long term reef monitoring.

- **Step 9: Continuous Scanning:** The device is programmed in such a way that it continues the scanning of the coral reef and repeat the above steps for some new damaged coral regions.

6. Attach drawing hand made/ computer made showing all the components of the invention.

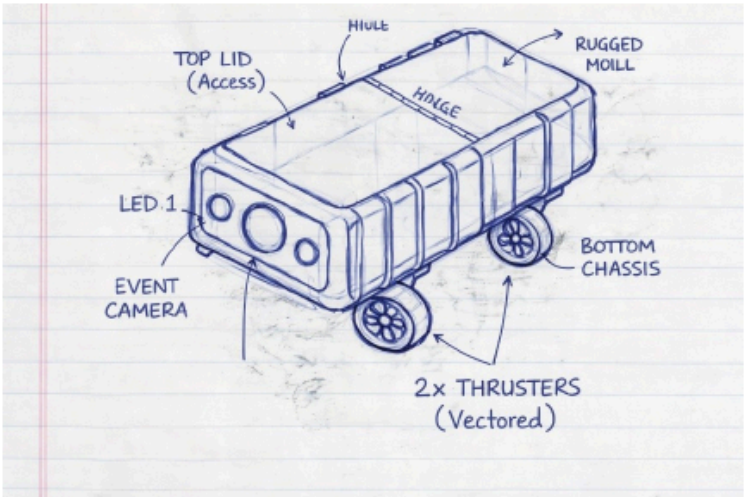

Figure 1: A rugged coral restoration device featuring a top access hinged lid, front mounted event driven camera with dual LEDs, and vectored thrusters for stable navigation and deployment.

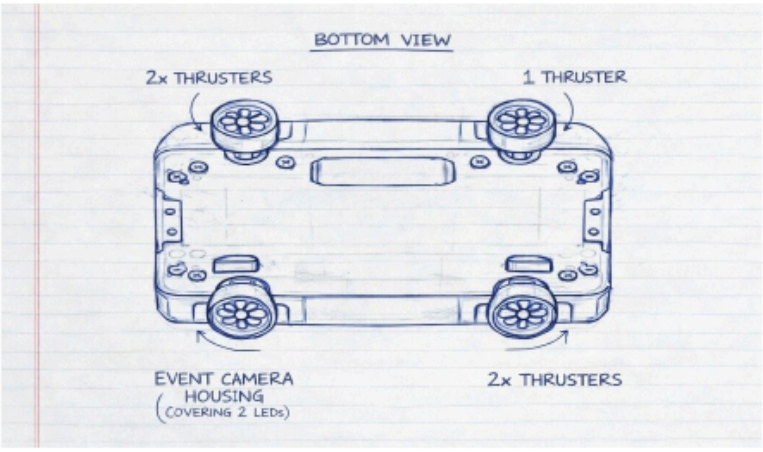

Figure 2: Bottom view showing four vectored thrusters and a protected event camera housing with dual LED coverage for underwater stability and vision.

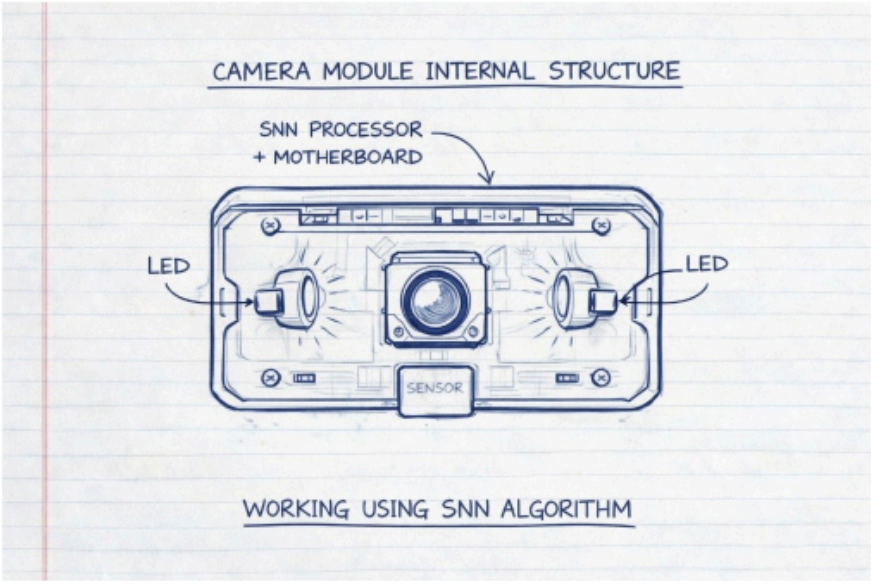

Figure 3: Internal camera module featuring a central event driven sensor with dual LED's and an integrated SNN processor for real time underwater reef analysis **.**

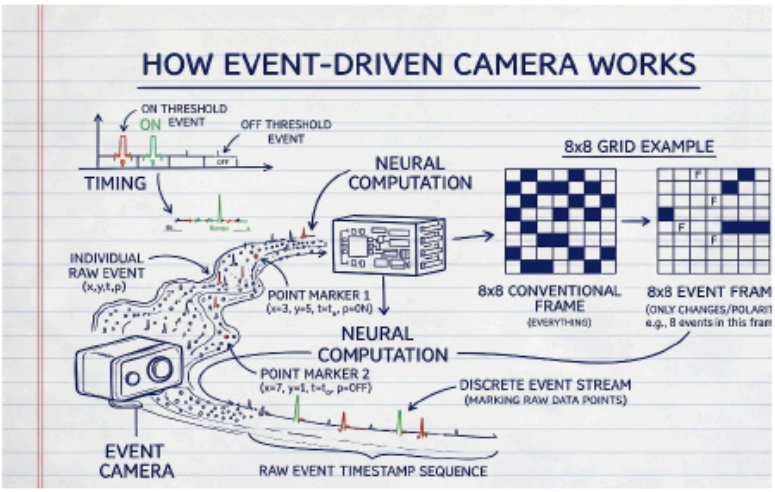

Figure 4: An **event driven camera** captures only changes in brightness at each pixel and sends them as fast, spike like signals instead of recording full image frames continuously.

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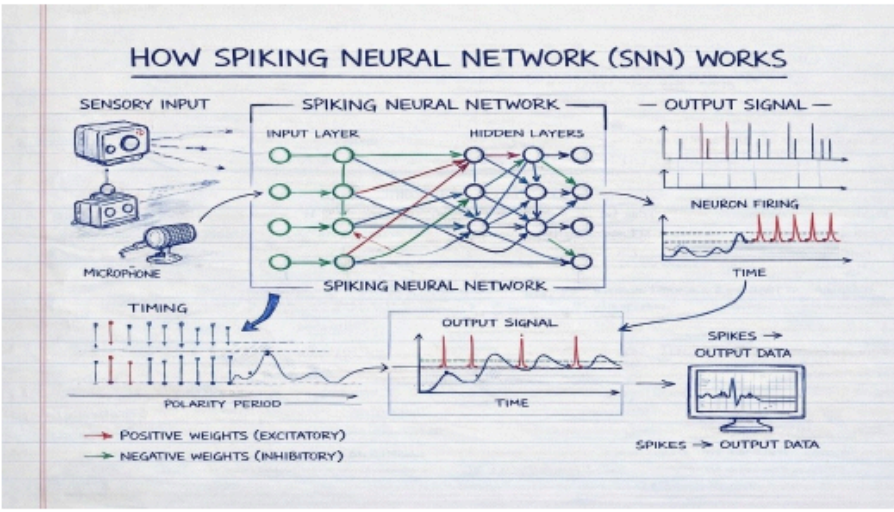


Figure 5 A Spiking Neural Network (SNN) processed the time based sensory inputs as discrete spikes, using excitatory and inhibitory neuron firing pattern to generate efficient and event driven output signals.

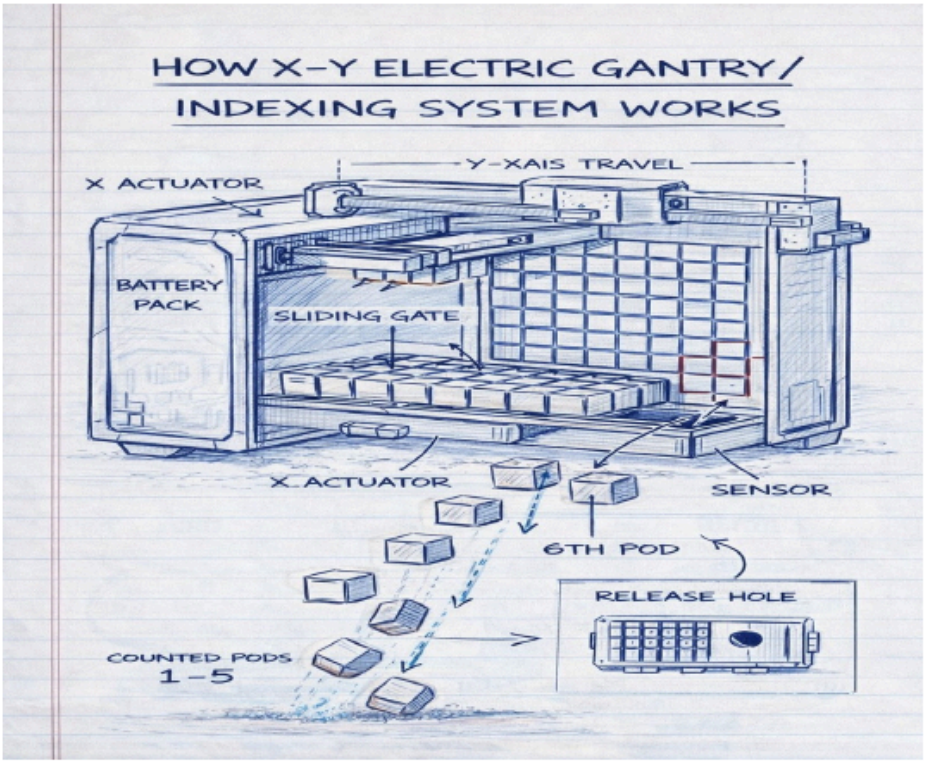


Figure 6 The XY electric gantry indexing system precisely moves across a pod grid using actuators and sensors to align and release selected coral pods through a controlled deployment hole.

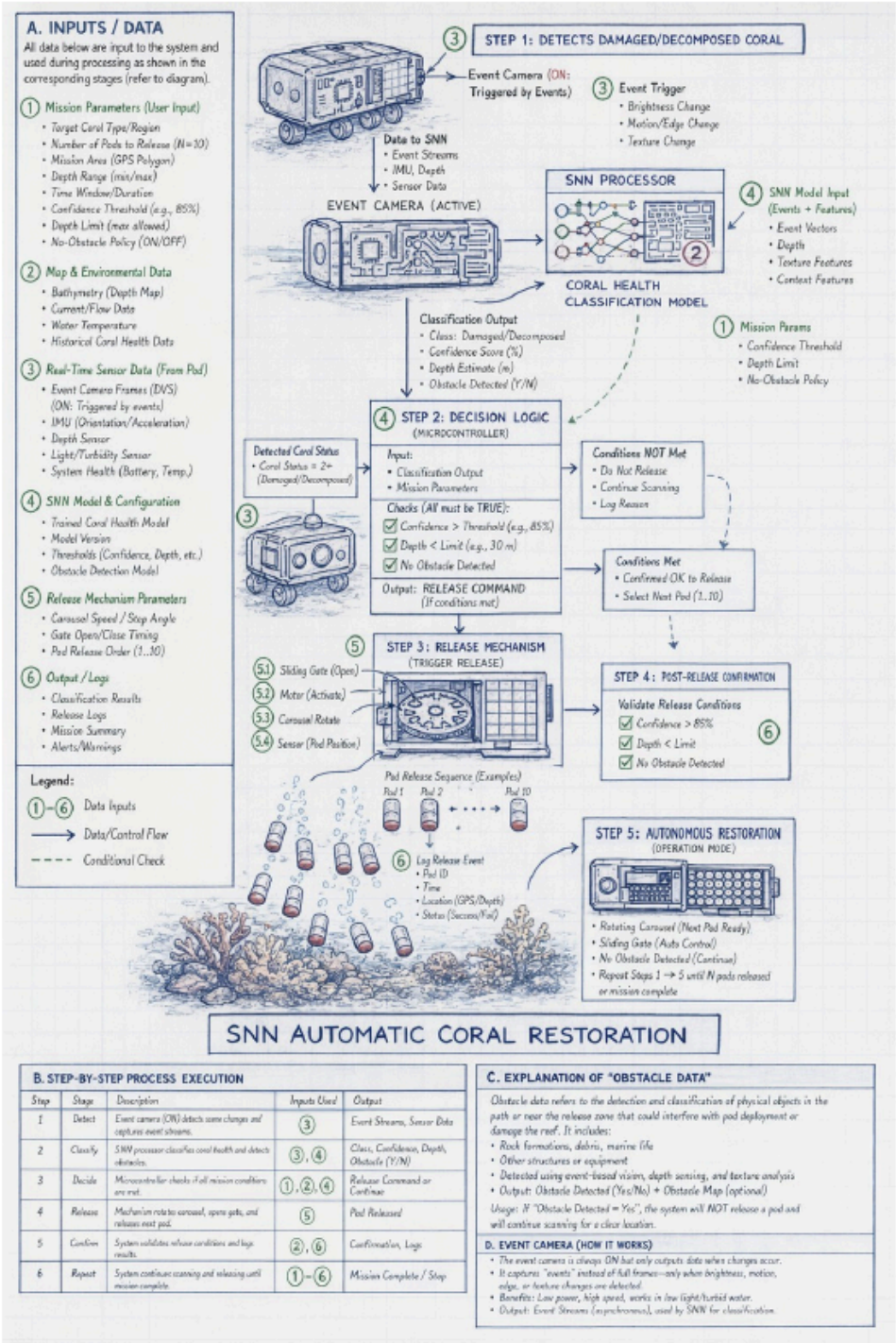
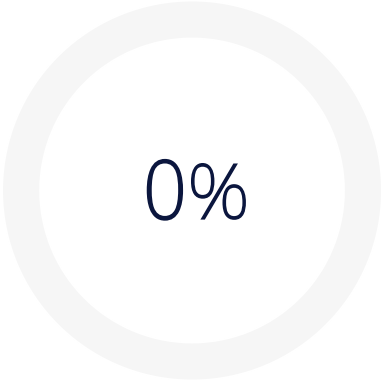


Figure 7 : An SNN based autonomous system detects damaged coral, validates environmental conditions and automatically deploys biodegradable pods through a controlled release mechanism for reef restoration.

Relevant Key phrases relating to your invention

S.NO.	Key phrases
1	Neuromorphic Event-Based Underwater Coral Detection System
2	Autonomous SNN-Based Real-Time Coral Health Classification
3	Adaptive XY Indexed Electric Gantry Pod Deployment Mechanism
4	Integrated Detection-to-Restoration Autonomous Marine Rehabilitation Platform

AI Content



	Text coverage	Words
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